Phase 1: AI Project Proposal
Email Content Summarization & Scheduler

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Project topic: Students struggle to schedule their timetables well through AIMS.
Target users/stakeholders: Students using AIMS, particularly students with large flexibility in course scheduling.

1 Problem Definition & Scope

1.1 Problem

The university students will receive a large volume of emails everyday, which contains assignment / quiz / exam announcement, event invitation, security alert, bank statement notification, promotion, etc. Spending time on checking these emails can be annoying. And the performance of filter function by email apps like Gmail and Outlook is not good enough. After checking these emails, make it to the schedule can also be another issues to the university students.

1.2 Scope

The project will contains classification of the emails, summarize an email from paragraph form to bullet-point form, integrate different emails to generate a schedule. But we will make an agent to help users to write emails.

2 AI Task Formulation

Task: Rank a set of timetables based on selected selection criteria, including, but not necessarily limited to, time of day, day of week, and consecutiveness

Inputs: List of occupied schedule slots / Slot in schedule's status as occupied / unoccupied (6 days of the week × 13 possible hours per day = 78 binary inputs)

Outputs: Scores based on various criteria (eg: "earliness", "lateness", "consecutiveness", etc; one output per criteria, potentially with an additional final layer for an overall aggregate score)

Constraints: Scores outputted from 0-100. Model will only be fed valid timetables (ie. without collisions)

Evaluation: Judge accuracy of suggestions; Scores should agree with the opinion of users.

Table 1: Comparison of candidate AI approaches.

Approach	Expected benefit	Main requirement	Main limitation
CNN	Learns timetable patterns	Labeled schedule ratings	Needs more data
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Data source: Survey students for their opinion on how a set of generated timetables should be scored. No privacy, consent, or access considerations needed.

3 Proposed AI Approaches

Present multiple plausible AI approaches. For each approach, briefly explain the core method, required data/resources, expected workflow, and practical feasibility. Divergent solutions are encouraged; do not present only one method under different names.

3.1 Convolutional NN

- Benefits
  - Better at preserving adjacency data
    - More accurate scores regarding "adjacency"
- Weaknesses
  - (to be written)
- Core method
  - Run a convolution kernel across the squares of the timetable before feeding the results through a simple fully connected neural network.
- Required resources
  - A laptop running Pytorch
- Required data
  - A set of randomly generated schedules and scores assigned by volunteers.

3.2 Linear regression

- Benefits
  - Trivial to optimize.
- Weaknesses
  - (to be written)
- Core method
  - (to be written)
- Required resources
  - (to be written)
- Required data
  - (to be written)

RLHF?

4 Comparative Analysis

A compact comparison table is encouraged when it saves space.

5 Expected Impact & Project Feasibility

Expected benefit: Easier scheduling, more efficient time management

Novelty: Likely very novel?

Value: High, helps students and the university schedule more efficiently, balance the preferences of students and professors

Data: Data of courses (scraped from AIMS Master Timetable), user preference data (for which timetables are scored higher)

Data collection plan: Survey students to select preferred timetables from generated timetables or score generated timetables. (eg. Ask students to score a set of 5 timetables from 0-100 based on the set of criteria listed above.)

Implementation: Create a dataset with scored timetables and optimise the regression/convolutional model to assign appropriate scores to schedules.

Tools: uhhh

Risks: Extremely minor: No identifying information needs to be collected (eg. name, class, major) and all collected information is not

Mitigation: N/A

Access to hardware: For Linear regression: Trivial (one laptop)

Access to software:

Access to data:

References

[1] Lance Le, "A Sample Title," *A sample journal*, 2023.

[2] Author Maple, "Rubik's Cube Methods," *Journal of Cube*, 2021.